

DELIVERABLE 5

Python Validation Pipeline Evidence

Pipeline 2 checks every plan the AI writes against rules taken from the documents, in plain Python. This document shows each check catching a real problem, with the validator's own output.

DELIVERABLE	5 of 18 - Python Validation Pipeline Evidence (SRS 1.10)
PIPELINE	Pipeline 2 - Python only, no Generative AI
DOCUMENT SET	Sitara Bank pack - 36 files, 422 clauses
PLANS GENERATED WITH	gpt-5.6-luna (live API)
CAPTURED ON	2026-09-28 10:57

Contents

- 1. Evidence at a glance 3
- 2. Requirement extraction 6
- 3. Requirement Matrix 8
- 4. Source validation 10
- 5. Coverage calculation 11
- 6. Traceability checking 12
- 7. Duplicate detection 13
- 8. Role relevance checking 14
- 9. Contradiction handling 15
- 10. Schema validation 17
- 11. Hallucination flagging 18
- 12. Automated tests and reproduction 19

1. Evidence at a glance

SRS Section 1.10, item 5 lists ten things the submission must demonstrate. Everything below was produced by running the project's real code (role_matrix/, python_validation/, hallucination_checks/, contradiction_checks/, schemas/) on the complete Sitara Bank document set.

Capture source: live API

The plans used as input in sections 4-11 were generated by **gpt-5.6-luna** on 2026-09-28 10:57. Pipeline 2 itself never uses AI.

228	100%	10/14	10	10
requirements extracted	match answer key	conflict cases solved	defects injected	defects caught

SRS item	Section	Proof in this document
Requirement extraction	2	228 requirements extracted by rules; compared field by field with the team's answer key
Requirement Matrix	3	215 active requirements across 12 roles; a sample of one role's matrix
Source validation	4	A citation to a section that does not exist, and to an adversarial document, are both rejected
Coverage calculation	5	Removing four modules drops mandatory coverage and lists every missing requirement
Traceability checking	6	Checklist and quiz items with broken citations lower the traceability score
Duplicate detection	7	Duplicates in the sources (vs answer key) and in a generated plan are both flagged
Role relevance checking	8	A requirement that belongs to another role is sent to manual review
Contradiction handling	9	Source conflicts vs answer key; overruled source and changed number caught in a plan; false skip reasons caught
Schema validation	10	Seven malformed outputs; six rejected with the exact field, one valid control accepted
Hallucination flagging	11	An invented rule and an invented quiz fact are flagged

How the demonstrations work

Sections 4 to 11 start from one real plan (PLN-EMP-002-20260928105839-a3d9, employee EMP-002, Customer Service Officer). For each check, one specific defect is written into a copy of that plan and the copy is validated again. Only what is **new** compared with the unmodified plan is shown, so each result can be tied to the defect that caused it.

Baseline plan	Value
Status	Unsupported
Mandatory coverage	12.4% (11 of 89)
Traceability	100.0%
Requirement consistency	97.7%
Row results	Verified with Warning: 1, Verified: 10, Requirement Missing: 78, Optional Not Included: 21

The baseline is not clean: Pipeline 2 already found these problems in what the model wrote. They are left in place and excluded from each demonstration below.

Requirement	Status	Validator explanation
POL-02-2.2	Verified with Warning	Due stage differs (matrix: First 90 Days).
CMP-01-2.5	Requirement Missing	Mandatory requirement from the matrix is not covered.
CMP-01-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.

Requirement	Status	Validator explanation
CMP-01-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
CMP-01-4.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
CMP-01-4.4	Requirement Missing	Mandatory requirement from the matrix is not covered.
CMP-01-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
CMP-01-5.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
FAQ-01-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
FAQ-01-8.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
FAQ-02-6.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-01-5.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-02-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-02-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-03-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-3.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-3.4	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-04-5.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-05-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-05-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-05-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-05-2.4	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-05-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-06-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-06-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-06-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-06-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-06-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.

Requirement	Status	Validator explanation
POL-07-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-3.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-07-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-08-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
POL-08-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-5.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-6.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
PRC-01-6.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
RD-01-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-3.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-3.4	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-01-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-02-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-02-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-02-2.4	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-02-4.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-02-4.2	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-03-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-03-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-03-5.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-05-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-05-2.3	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-05-3.1	Requirement Missing	Mandatory requirement from the matrix is not covered.
SOP-06-2.1	Requirement Missing	Mandatory requirement from the matrix is not covered.

2. Requirement extraction

Requirements are extracted from document clauses by rules in `role_matrix/extractor.py` and `config/requirement_rules.yaml`. No AI is involved, so the same documents always give the same matrix.

Attribute	Rule
Obligation	must / shall / required / mandatory = Mandatory; should = Recommended; may = Optional
Requirement type	From the verb: complete, acknowledge, know, perform, report ...
Applicable roles	From the section's 'Applies to' line; role names in the clause or document title as fallback
Due stage	From the deadline wording (Day 1, first week, within N days, end of probation); category default when none
Priority	From obligation, document criticality (security, privacy, AML, fraud, compliance) and stage
Assessment	From the requirement type (knowledge quiz, practical, signed acknowledgement ...)
Prerequisites	'X must be completed before Y' clauses are linked

Extraction results

Obligation	Requirements
Mandatory	190
Recommended	22
Optional	16

Examples of extracted requirements

ID	Clause	Extracted attributes
POL-01-2.1	Every new employee must sign the Employee Handbook acknowledgement form on Day 1.	Mandatory · Must Acknowledge · Day 1 · High priority · Signed Acknowledgement Roles: All roles
POL-04-3.2	Employees must change their passwords every 60 days.	Mandatory · Must Know · Week 1 · High priority · Knowledge Quiz Roles: All roles
POL-01-5.2	Employees may use the staff cafeteria and prayer rooms during scheduled breaks.	Optional · Optional · Week 1 · Low priority · None Roles: All roles
SOP-04-3.1	The vault must be opened only under dual control by two authorised staff members.	Mandatory · Must Know · Week 2 · Medium priority · Knowledge Quiz Roles: Branch Manager, Operations Coordinator
PRC-01-2.3	New joiners must attend the welcome orientation session on Day 1.	Mandatory · Must Complete · Day 1 · High priority · Knowledge Quiz Roles: All roles
POL-03-2.2	Employees must apply for annual leave at least 14 days in advance through the HR portal.	Mandatory · Must Know · Week 1 · Medium priority · Knowledge Quiz Roles: All roles
FAQ-01-2.1	You should apply for annual leave at least 7 days in advance.	Recommended · Recommended · Week 2 · Low priority · None Roles: All roles Inactive: overruled by POL-03-2.2

Accuracy against the team's answer key

The dataset includes a hand-made answer key (`requirements_master.csv`, 228 requirements) that was written independently of the code. Every extracted requirement was compared with it field by field:

Field	Agree	Agreement
Requirement found	228 / 228	100.0%
Obligation	228 / 228	100.0%
Due stage	228 / 228	100.0%
Priority	228 / 228	100.0%
Applicable roles	228 / 228	100.0%
Active / inactive	220 / 228	96.5%

The 8 disagreements, and why

Requirement	Field	Answer key	Application	Reason
SOP-03-2.2	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
FAQ-01-7.1	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
FAQ-01-8.1	active	False	True	Should have been overruled by a policy (conflict case C09, not detected - see section 9).
FAQ-02-4.1	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
CMP-01-2.1	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
CMP-01-2.2	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
CMP-01-2.3	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.
CMP-01-2.4	active	True	False	Duplicate of a higher-precedence clause. The application keeps one copy active; the answer key keeps both.

3. Requirement Matrix

The Role Requirement Matrix is the Python ground truth. Of 228 extracted requirements, **215** are active (180 mandatory, 137 role-specific). 13 were deactivated because a higher-precedence source overrules them or they duplicate another clause - they stay in the matrix for traceability but are never expected in a plan.

Role	Active requirements	Mandatory	Recommended	Optional	Role-specific
Branch Manager	125	104	11	10	47
Customer Service Officer	110	89	12	9	32
Sales Executive	102	83	11	8	24
HR Executive	84	67	9	8	6
Finance Associate	89	71	10	8	11
Operations Coordinator	102	85	9	8	24
IT Support Engineer	90	73	8	9	12
Compliance Officer	97	79	9	9	19
Marketing Executive	91	73	9	9	13
Data Analyst	92	74	9	9	14
Cash Teller	110	91	11	8	32
Team Leader	93	75	10	8	15

Sample: the first rows of the Customer Service Officer matrix

ID	Requirement	Obligation	Stage	Priority	Assessment
POL-01-2.1	Every new employee must sign the Employee Handbook acknowledgement form on Day 1.	Mandatory	Day 1	High	Signed Acknowledgement
POL-01-2.3	New employees must collect their employee ID card from the HR desk on Day 1.	Mandatory	Day 1	High	Knowledge Quiz
POL-04-2.2	All employees must sign the Acceptable Use acknowledgement on Day 1.	Mandatory	Day 1	High	Signed Acknowledgement
POL-05-2.4	Employees must acknowledge the Code of Conduct on Day 1.	Mandatory	Day 1	High	Signed Acknowledgement
PRC-01-2.1	New joiners must report to the HR desk at 9:00 AM on Day 1.	Mandatory	Day 1	High	Knowledge Quiz
PRC-01-2.3	New joiners must attend the welcome orientation session on Day 1.	Mandatory	Day 1	High	Knowledge Quiz
PRC-01-6.1	Branch new joiners must complete a branch security walkthrough on Day 1.	Mandatory	Day 1	High	Knowledge Quiz
POL-01-2.2	New employees must submit copies of their educational certificates to HR within the first week of joining.	Mandatory	Week 1	Medium	Knowledge Quiz
POL-01-2.4	New employees should meet their reporting manager within the first week of joining to agree on role expectations.	Recommended	Week 1	Low	None
POL-01-3.2	Employees must record attendance using the biometric system at the start and end of each working day.	Mandatory	Week 1	Medium	Knowledge Quiz
POL-01-3.3	Employees may request flexible timing with written approval from their reporting manager.	Optional	Week 1	Low	None
POL-01-4.1	Employees must follow the business formal dress code on all customer-facing days.	Mandatory	Week 1	Medium	Knowledge Quiz
POL-01-4.2	Employees should refer to the Workplace Conduct Policy (POL-05) for detailed behavioural standards.	Recommended	Week 1	Low	None
POL-01-5.1	Employees must use bank equipment only for official purposes.	Mandatory	Week 1	Medium	Knowledge Quiz

Deactivated requirements

Requirement	Reason
CMP-01-2.1	Duplicate of POL-04-2.1
CMP-01-2.2	Duplicate of POL-06-2.2
CMP-01-2.3	Duplicate of POL-08-2.1
CMP-01-2.4	Duplicate of POL-05-3.3
CMP-01-3.1	Overruled by POL-07-2.1
FAQ-01-2.1	Overruled by POL-03-2.2
FAQ-01-7.1	Duplicate of POL-03-3.2
FAQ-02-1.1	Overruled by POL-04-3.2
FAQ-02-2.1	Overruled by POL-04-4.1
FAQ-02-4.1	Duplicate of POL-04-5.1
RD-01-3.3	Overruled by SOP-03-3.1
SOP-01-3.3	Overruled by SOP-02-2.2
SOP-03-2.2	Duplicate of POL-07-3.1

4. Source validation

Every cited document and section is looked up in the stored clauses (*hallucination_checks/support.py*). A citation passes only if the section exists in the Active, trusted, non-quarantined version.

Defect written into the plan	M01: cited POL-02 §2.2 changed to §9.9 (does not exist)
Status after validation	Unsupported
Mandatory coverage	12.4% → 11.2% (-1.2%)
Traceability	100.0% → 98.2% (-1.8%)

New requirement-level results

Requirement	Status	Validator explanation
POL-02-9.9	Source Support Missing	Cited document/section does not exist.
POL-02-2.2	Requirement Missing	Mandatory requirement from the matrix is not covered.

New item-level findings

Check	Severity	Where	Message
Traceability	High	M01 requirement [POL-02 §9.9]	Source is missing.

Second case: citing an adversarial document

Defect written into the plan	M01: added a requirement citing ADV-01 §1.1 (an adversarial, untrusted document)
Result	ADV-01-1.1: Unsupported Requirement - Cited source is untrusted or quarantined (adversarial).

Source check result	Meaning	Row status
valid	Active, trusted section exists	compared with the matrix
missing	No such document or section	Source Support Missing
outdated	Section exists only in a superseded version	Outdated Source
untrusted / quarantined	Adversarial document or quarantined clause	Unsupported Requirement

5. Coverage calculation

Coverage = covered mandatory requirements / mandatory requirements expected for the role x 100. A requirement counts only if its citation is valid, not overruled, numerically faithful and supported by the source text.

Defect written into the plan	removed the first 4 modules (7 requirements)
Status after validation	Incomplete
Mandatory coverage	12.4% → 4.5% (-7.9%)
Traceability	100.0% → 100.0%

New item-level findings

Check	Severity	Where	Message
Prerequisite	Medium	M05	Prerequisite module M04 does not exist.

Requirements reported as missing

Requirement	Python expected	Due stage	Status
POL-02-2.2	Employees must complete all mandatory onboarding modules before the end of probation.	First 90 Days	Requirement Missing
POL-02-2.4	Employees must complete a probation review meeting with their reporting manager within 90 days of joining.	First 90 Days	Requirement Missing
POL-03-2.3	Employees must not take more than 10 consecutive working days of annual leave without Department Head approval.	Week 1	Requirement Missing
POL-03-3.1	Employees must inform their reporting manager before 10:00 AM on any day they cannot come to work due to sickness.	Week 1	Requirement Missing
POL-03-3.2	Sick leave of more than 2 consecutive days must be supported by a medical certificate.	Week 1	Requirement Missing
POL-05-4.1	Employees must declare any conflict of interest to HR within the first week of joining.	Week 1	Requirement Missing
POL-05-5.1	Employees must not post confidential bank information on social media.	Week 1	Requirement Missing

Arithmetic: 4 covered / 89 expected x 100 = **4.5%**. The plan status becomes **Incomplete**; below 100% the Approve button is blocked and the reviewer uses **Add missing requirements**.

6. Traceability checking

Every checklist item, task and quiz question - not only the covered requirements - must cite a valid source.
Traceability = items with a valid source / all items x 100.

Defect written into the plan	M01: a checklist item cites a non-existent section, a quiz question cites a non-existent document
Status after validation	Unsupported
Mandatory coverage	12.4% → 12.4%
Traceability	100.0% → 96.4% (-3.6%)

New item-level findings

Check	Severity	Where	Message
Traceability	Medium	M01 checklist [POL-02 §7.7]	Source is missing.
Traceability	Medium	M01 quiz [POL-99 §2.2]	Source is missing.

7. Duplicate detection

Duplicates are caught at two levels: between source clauses when the matrix is built, and inside a generated plan.

Defect written into the plan	POL-02-2.2 added to M02 as well; module M01 copied as M99
Status after validation	Unsupported
Mandatory coverage	12.4% → 12.4%
Traceability	100.0% → 100.0%

New item-level findings

Check	Severity	Where	Message
Duplicate learning content	Low	M02/POL-02-2.2	POL-02-2.2 is covered in more than one module (M01 and M02).
Duplicate learning content	Low	M99/POL-02-2.2	POL-02-2.2 is covered in more than one module (M01 and M99).
Duplicate learning content	Medium	M01/M99	Modules are near-duplicates.
Duplicate learning content	Low	M01-Q1/M99-Q1	Quiz questions are near-duplicates.
Duplicate learning content	Low	M01-Q2/M99-Q2	Quiz questions are near-duplicates.
Duplicate learning content	Low	M01-T1/M99-T1	Tasks are near-duplicates.
Duplicate learning content	Low	M01-T2/M99-T2	Tasks are near-duplicates.

Source-level duplicates vs the answer key

Requirement	Expected duplicate of	Detected duplicate of	Correct
SOP-03-2.2	POL-07-3.1	POL-07-3.1	Yes
RD-01-12.3	POL-02-6.3	not detected	No
CMP-01-2.1	POL-04-2.1	POL-04-2.1	Yes
CMP-01-2.2	POL-06-2.2	POL-06-2.2	Yes
CMP-01-2.3	POL-08-2.1	POL-08-2.1	Yes
CMP-01-2.4	POL-05-3.3	POL-05-3.3	Yes
FAQ-01-7.1	POL-03-3.2	POL-03-3.2	Yes
FAQ-02-4.1	POL-04-5.1	POL-04-5.1	Yes

Plan-level similarity threshold: 0.85 (config/validation_rules.yaml).

8. Role relevance checking

A requirement can be perfectly valid and still wrong for this employee. Anything outside the role's matrix is not counted as coverage and is routed to a reviewer.

Defect written into the plan	M01: added POL-02-5.1 (HR Executive only) to a Customer Service Officer plan
Status after validation	Unsupported
Mandatory coverage	12.4% → 12.4%
Traceability	100.0% → 100.0%

New requirement-level results

Requirement	Status	Validator explanation
POL-02-5.1	Manual Review Required	Valid requirement but not applicable to the role Customer Service Officer.

Clause added: *HR Executives must verify the educational certificates of every new joiner within 14 calendar days of the new joiner's start date.*

9. Contradiction handling

Contradictions are handled at two levels: conflicting source clauses are resolved by the precedence order when the matrix is built, and a plan that cites an overruled clause or changes a number is flagged.

Defect written into the plan	added CMP-01-3.1 (overruled by POL-07-2.1); POL-02-2.4: 'Employees must complete a probation review meeting with their reporting manager within 90 days of joining.' changed to 'Employees must complete a probation review meeting with their reporting manager within 97 days of joining.'
Status after validation	Contradictory
Mandatory coverage	12.4% → 11.2% (-1.2%)
Traceability	100.0% → 100.0%

New requirement-level results

Requirement	Status	Validator explanation
CMP-01-3.1	Contradiction Detected	Source is overridden by POL-07-2.1 (precedence rules).
POL-02-2.4	Contradiction Detected	Numbers 97 do not appear in the cited source.

Invalid 'overruled by' claims

The model may skip a clause only if a higher-authority document overrules it. Python checks every claim:

Claim written by the model	Validator finding
POL-02-3.1 - overruled claim	The model skipped POL-02-3.1 as 'overruled by POL-02-2.3', but a document cannot overrule its own rule.
POL-04-3.2 - overruled claim	The model skipped POL-04-3.2 as 'overruled by FAQ-02-1.1', but FAQ-02 (FAQ) does not outrank POL-04 (Policy).

Source conflicts vs the answer key: 10 of 14 handled correctly

Case	Type	Expected	Application result	Rule applied	Correct
C01	Policy version change + outdated FAQ	POL-03 v2.0 §2.2	POL-03-2.2	Policy (level 1) outranks FAQ (level 7)	Yes
C02	FAQ contradicts policy	POL-02 v2.0 §2.3	POL-02-2.3	Policy (level 1) outranks FAQ (level 7)	Yes
C03	FAQ contradicts policy	POL-04 v2.0 §3.2	POL-04-3.2	Policy (level 1) outranks FAQ (level 7)	Yes
C04	SOP contradicts SOP	SOP-02 v2.0 §2.2	SOP-02-2.2	Same level: later effective date wins (2026-06-15 > 2025-05-01)	Yes
C05	Role description contradicts SOP	SOP-03 v2.0 §3.1	SOP-03-3.1	SOP (level 3) outranks Role Description (level 5)	Yes
C06	Handbook contradicts policy	POL-02 v2.0 §3.1	POL-02-3.1	Policy (level 1) outranks Handbook (level 4)	Yes
C07	FAQ contradicts policy	POL-04 v2.0 §4.1	POL-04-4.1	Policy (level 1) outranks FAQ (level 7)	Yes
C08	Compliance document contradicts newer-dated policy	POL-07 v2.0 §2.1	POL-07-2.1	Policy (level 1) outranks Compliance (level 2)	Yes
C09	FAQ contradicts policy deadline	POL-02 v2.0 §2.2	not detected		No
C10	Ambiguous timing	SOP-05 v2.0 §2.2	not detected		No
C11	Missing referenced document	Flag missing source	Missing reference: MKT-GD-01	Referenced document is not in the pack	Yes
C12	Missing referenced document	None	not detected		No
C13	General vs specific rule (not a conflict)	Both valid	Both valid	A general rule and a role-specific rule apply to different scopes.	Yes
C14	Sequencing ambiguity	PRC-01 §6.2 (safety)	not detected		No

Not detected cases are a known limitation of rule-based detection: they need reasoning without matching numbers or opposite wording (for example '60 days' against 'end of probation'). They are listed in the README's limitations.

10. Schema validation

Every GenAI response is validated against the Pydantic schema in `schemas/plan_schema.py` before anything else. The same real plan was broken in six different ways; one unchanged copy was included as a control.

Case	Result	Field	Pydantic message
Valid plan returned by the model	Accepted	-	Valid
Module category outside the allowed list	Rejected	modules.0.category	Input should be 'Policy', 'Compliance', 'Process', 'Role Skills' or 'Orientation'
Module with no covered requirement	Rejected	modules.0.requirements_covered	List should have at least 1 item after validation, not 0
Checklist item without a source	Rejected	modules.0.checklist.0.source_document_id	Field required
Quiz type not supported	Rejected	modules.0.quiz.0.type	Input should be 'multiple_choice', 'multiple_response', 'true_false' or 'scenario'
Duration of 2,000 minutes	Rejected	modules.0.estimated_duration_minutes	Input should be less than or equal to 600
Required field 'modules' missing	Rejected	modules	Field required

A rejected response is never stored. It is retried with the Pydantic message appended to the prompt (see Deliverable 4, section 11).

11. Hallucination flagging

A generated statement must be supported by the clause it cites. Text similarity below 0.18 (requirements) or 0.12 (tasks, quizzes), or a number that does not appear in the source, is flagged.

Defect written into the plan	POL-02-2.2: statement replaced with an invented benefit; quiz M01-Q1 replaced with an invented fact
Status after validation	Unsupported
Mandatory coverage	12.4% → 11.2% (-1.2%)
Traceability	100.0% → 100.0%

New requirement-level results

Requirement	Status	Validator explanation
POL-02-2.2	Unsupported Requirement	Low similarity to the cited source (0.0) - possible hallucination.

New item-level findings

Check	Severity	Where	Message
Possible hallucination	Medium	M01 quiz [POL-02 §2.2]	Question is weakly related to its cited source (similarity 0.0).
Unsupported fact	High	M01 quiz [POL-02 §2.2]	Correct answer uses 25, which is not in the source.

Other checks in the validator

Check	Finding when a defect was injected
Quiz answer	Correct answer is not one of the options.
Quiz distractor	Distractor 'Before the end of probation' is literally stated in the source.
Security scan of the plan	Instruction-like text in a generated plan is flagged High
Stage spread	Warning if more than 40% of modules are on Day 1
Rubric weights	Practical rubrics must add up to 100

12. Automated tests and reproduction

The same checks are asserted by the project's test suite on every run - result while building this document: **5 failed, 12 passed, 36 deselected in 12.34s**.

Test	Proves
test_matrix_meets_dataset_minimums	Extraction meets the SRS dataset minimums
test_faq_conflict_resolved_by_policy	FAQ loses to policy
test_compliance_conflict_uses_precedence	Precedence order applied
test_scope_specific_rule_is_not_a_conflict	General vs role-specific rule both kept
test_duplicates_detected	Source duplicates found
test_missing_reference_detected	Missing referenced document found
test_invalid_skip_reasons_are_detected	False 'overruled by' claims rejected
test_schema_rejects_bad_json	Schema rejects malformed plans
test_validator_catches_injected_errors	Hallucination, wrong number, untrusted source, missing requirement caught
test_quiz_and_sequence_errors_flagged	Quiz answer and sequencing errors caught
test_approval_needs_full_coverage_and_missing_items_can_be_added	Coverage gate before approval

REPRODUCE

```
python -m src.evidence.build_validation_evidence          # offline stand-in for plan generation
python -m src.evidence.build_validation_evidence --live   # real API from .env
python -m pytest -q                                       # the full test suite
```